

Fantasy Football Tracker — technical reference

Personal tracker: **Python collector** POSTs snapshots into **Spring Boot 3.5 + MySQL**, and **Angular 21** GETs a homepage + dashboard. The frontend never writes. Gameweeks are **rows**, not tables.

Sources

source	Slug example	How data arrives
sofascore	premier-league, laliga, nations-league, ...	Logged-in SofaScore Fantasy XHRs + <code>SOFASCORE_SESSION</code>
fpl	premier-league-fantasy	Public FPL JSON (<code>FPL_ENTRY_ID</code>)
laliga-fantasy	laliga-fantasy-official	Excel workbook (<code>import-excel</code>)
wsl	wsl-fantasy	Official my-team XHR (<code>x-game-token</code>) + public feeds

Constraints that shaped the code: collector POSTs / frontend only GETs; MySQL on 3307; never commit `.env`, cookies, or tokens; do not invent SofaScore API paths or scrape HTML; do not reverse official LaLiga/WSL apps; one process on port 8080.

Architecture

```
collector (pull / import / import-excel)
  POST /api/v1/ingest/snapshots
  POST /api/v1/ingest/transfers
    → IngestService upserts entities
frontend GET /api/v1/home
  → HomePageService (brands + Players of the Week)
frontend GET /api/v1/competitions...
  → CompetitionQueryService + branding/crests/portraits
```

Captain **display** is always $\times 2$ ($\times 3$ if triple captain). Ingest stores **raw** points. Official WSL `totalPoints` already includes $\times 2$, so the collector divides the captain's value by 2 before POST.

Club vs national team: the same SofaScore player id can appear as a club side (e.g. FC Barcelona in LaLiga) and a national side (Spain in Nations League). Club name, crest id, and shirt number for display are stored on `squad_pick` (competition context), not only on the shared `player` row.

Backend (Spring Boot)

Package: `com.fantasytracker.backend`. Profile `mysql` uses XAMPP MariaDB on 3307 (`fantasy_tracker`). Default profile is H2 + seed JSON for tests.

BackendApplication

- **main** — boots Spring. Created as the single process that owns port 8080.

Config

WebConfig.addCorsMappings

Allows the Angular origin (`fantasy.cors.allowed-origins`, default `http://localhost:4200`) on `/api/**`. Exists so the dashboard can GET localhost:8080 from localhost:4200.

SeedRunner.run

`ApplicationRunner` (on unless `fantasy.seed.enabled=false`). If the competition table is empty, ingests `seed/pl-gw1.json` and `seed/pl-gw2.json` so H2/demo boots with "Sofa Saints". Skipped on a populated MySQL database.

HTTP — ingest (`IngestController`)

Optional header `X-Ingest-Token` when `fantasy.ingest.token` is set.

Method	Path	Why
ingest	POST <code>/api/v1/ingest/snapshots</code>	Collector (and Excel/FPL/WSL adapters) upsert one gameweek squad + scores.
ingestTransfers	POST <code>/api/v1/ingest/transfers</code>	Official transfer rows (paired in/out, or unpaired LaLiga buys/sells).
cleanupSeed	POST <code>/api/v1/ingest/cleanup-seed</code>	Removes players whose <code>externalId</code> starts with <code>seed-</code> after switching to live data.

HTTP — homepage (`HomeController`)

Method	Path	Why
home	GET <code>/api/v1/home</code>	League brand wash + Players of the Week for the landing page.

HTTP — read (`CompetitionController`)

Frontend-only surface for the leagues dashboard.

Method	Path	Why
list	GET <code>/api/v1/competitions</code>	Dropdown: name, season, branding, gameweek numbers.
team	GET <code>/api/v1/competitions/{id}/team?gameweek=</code>	Squad for a week (latest if omitted).
gameweek	GET <code>/api/v1/competitions/{id}/gameweeks/{n}</code>	Same payload keyed by week number.

Method	Path	Why
compare	GET /api/v1/competitions/{id}/compare?from=&to=	Week-vs-week player deltas.
totals	GET /api/v1/competitions/{id}/totals	Season sum of team week scores.

HomePageService

Why it exists: one read model for the homepage so Angular does not assemble brands/POTW from many competition calls.

- **homePage** — all competitions; default prefers SofaScore **premier-league**; unique **LeagueBrands** keyed by **CompetitionBranding.brandKey**; one **PlayerOfTheWeek** per competition that has a scored squad.
- **playerOfTheWeek(Competition)** — resolves latest scored gameweek; picks the squad row with max **CaptainScoring.effective(raw, captain, tripleCaptain)**; club / shirt / crest prefer **pick-scoped** fields, then fall back to **Player**.
- **resolveLatestScoredGameweek** — prefer latest **status=finished** week with picks; else latest team score with points > 0; else last score row.
- **displayBrandName** — short labels for the brand wash ("10783" → "Nations League", etc.).

Players of the Week vs MVP

Concept	Where	Rule
POTW	Backend HomePageService	Per competition: highest captain-effective points in that competition's resolved latest GW (your squad only).
MVP	Frontend HomeComponent.topPlayerOfTheWeek	Highest points across the returned playersOfTheWeek list (cross-competition).
GW card	Frontend DashboardComponent.playerOfTheWeek	Highest pick.points in the selected competition gameweek.
Season high / low GW	Frontend from totals	Max / min team points across scored weeks (cards between squad and Compare).
LaLiga buy / sale cards	transferMarket.mostExpensivePurchase / highestSale	Season max priceIn / priceOut across official transfer rows.

IngestService

Why it exists: one upsert path for SofaScore, FPL, WSL, and Excel so the dashboard schema stays stable.

- **upsertSnapshot** — competition, gameweek, team, picks, player scores; drops stale picks for that week; team points from the request or from starter points after **CaptainScoring.effective**.
- **upsertTransfers** — replaces that week's transfer rows; applies **transferPenalty** onto **TeamGameweekScore**.

- **removeSeedPlayers** — cleanup for **seed-*** ids.
- **findAllCompetitions** — used by tests/tooling.
- **upsertCompetition** / **upsertGameweek** / **upsertTeam** / **upsertPlayer** / **upsertPick** / **upsertPlayerScore** / **upsertTeamScore** — idempotent writes keyed by source+external id (or team+player+week).
- **upsertPick** — also writes **pick-scoped** **club**, **clubExternalId**, **shirtNumber**, plus **injured** / **suspended** from the snapshot payload.
- **upsertGameweekForTransfers** / **applyTransferPenalty** — transfers can arrive for a week that has no squad yet.
- **removeStalePicks** — a new XI replaces last week's leftover rows.
- **toJson** / **blankToNull** — stores pick **breakdown** as JSON text; normalises empty club ids.

CompetitionQueryService

Why it exists: GET payloads include derived media (crest, portrait, colours) and captain **display** points without the frontend knowing chips.

- **listCompetitions** — all competitions with branding + distinct gameweek numbers.
- **teamView** / **gameweekView** — same builder; **teamView** picks latest week if unspecified.
- **compare** — players in either week; points use captain multiplier per week's triple-captain flag; club via **clubOf**.
- **totals** — ordered week scores + season total.
- **buildTeamView** — starters/bench, portraits, crests (pick-scoped club), transfers, transfer-market summary (incl. LaLiga season extremes).
- **buildTransfers** — stored **gameweek_transfer** rows if present; otherwise **squad-diff** against the previous week (WSL and gaps in official feeds).
- **toTransfer (overloads)** — maps a player + price + counterpart into the dashboard transfer card; the **SquadPick** overload uses pick-scoped club.
- **clubOf** / **clubExternalIdOf** / **shirtNumberOf** — prefer pick fields, else **Player** (legacy rows before migration).
- **buildTransferMarket** / **summarizeMarket** / **toCounterpartGroups** / **Accumulator** — LaLiga market vs release-clause totals (week + season).
- **findMostExpensivePurchase** / **findHighestSale** / **toHighlight** — season-wide max buy / max sale from **gameweek_transfer** prices for POTW-style cards on Official LaLiga.
- **previousGameweek** / **tripleCaptain** / **pickComparator** / **positionOrder** — GK→DEF→MID→FWD, starters before bench.
- **indexPicks** / **indexScores** / **pointsOf** / **resolveGameweek** / **requireCompetition** / **requireTeam** / **requireGameweek** — lookups that 404 as 404-style failures.

Scoring and media helpers

CaptainScoring

- **multiplier** — 1, 2, or 3.
- **effective** — raw × multiplier. Created so SofaScore/FPL/WSL all store raw and display chips the same way.

CompetitionBranding

Maps **source** + **externalId** onto SofaScore unique-tournament ids. Official FPL / LaLiga Fantasy / WSL reuse those ids even though **source** is not **sofascore**.

Public

Method	Purpose
logoUrl / logoDarkUrl	Tournament image (homepage prefers dark).
flagUrl / countryName	Category flag + label.
primaryColor / secondaryColor	CSS brand colours for wash / cards / headers.
brandKey	Shared visual key so FPL↔PL and Official LaLiga↔SofaScore LaLiga share one logo slot.

Private: **brandingTournamentId** (remaps **laliga-fantasy**→8, **fpl**→17, **ws1**→1044; else SofaScore numeric id), **numericId**, **country**, **colors**.

Brand keys

Key	League	Primary / secondary
17	Premier League	#3c1c5a / #f80158
8	LaLiga	#2f4a89 / #f4a32e
23	Serie A	#09519e / #008fd7
34	Ligue 1	#091c3e / #a9c011
35	Bundesliga	#e2080e / #8e0902
7	Champions League	#062b5c / #086aab
679	Europa League	#3d1a08 / #f37d25
10783	Nations League	#3a4179 / #e5a422
242	MLS	#e2231a / #062f69
325	Brasileirão	#C7FF00 / #969696
1044	WSL	#06121e / #00c2cb

ClubCrests.url

FPL uses Premier League badge CDN `t{code}.png`. Everyone else uses `img.sofascore.com/api/v1/team/{id}/image`, with name fallbacks for PL/LaLiga clubs when the id is missing. WSL is **not** mapped through men's PL names.

PlayerPortraits.url

Numeric SofaScore player id → `img.sofascore.com/api/v1/player/{id}/image`. Non-numeric ids (unresolved WSL `wsl-...`, FPL `fpl-...`) get no portrait.

TransferChannels

- **isMarket / channel** — blank/"Market" vs any other counterpart = release clause. Exists for official LaLiga Excel deals.

Entities (JPA)

Lombok `@Data`. Tables match `sql/schema.sql`. Migrations: `sql/migrate-squad-pick-club-context.sql`, `sql/migrate-squad-pick-suspended.sql` (also applied via `spring.jpa.hibernate.ddl-auto=update` locally).

Entity	Role
Competition	Unique (<code>source</code> , <code>externalId</code> , <code>season</code>).
Gameweek	Week row (<code>number</code> , <code>status</code> , <code>dates</code>).
FantasyTeam	One team per competition.
Player	Unique (<code>source</code> , <code>externalId</code>); shared identity (<code>name</code> , <code>portrait id</code>). Club fields are a soft cache / transfer fallback.
SquadPick	Role, captain/VC, price, injured , suspended ; plus pick-scoped club , clubExternalId , shirtNumber .
PlayerGameweekScore	Raw <code>points</code> , optional <code>rating</code> + <code>breakdown</code> .
TeamGameweekScore	Week total, triple-captain flag, transfer penalty.
GameweekTransfer	Nullable in/out players, prices, counterpart, channel, sort order.

Repositories

Spring Data query methods — no custom SQL.

- `CompetitionRepository.findBySourceAndExternalIdAndSeason`
- `GameweekRepository.findByCompetition_IdAndNumber`,
`findByCompetition_IdOrderByNumberAsc`
- `FantasyTeamRepository.findByCompetition_Id`
- `PlayerRepository.findBySourceAndExternalId`, `findByExternalIdStartingWith` (seed cleanup)
- `SquadPickRepository.findByFantasyTeam_IdAndGameweek_Id`,
`findByFantasyTeam_IdAndPlayer_IdAndGameweek_Id`, `deleteByPlayer_Id`
- `PlayerGameweekScoreRepository.findByPlayer_IdAndGameweek_Id`,
`findByGameweek_IdAndPlayer_IdIn`, `deleteByPlayer_Id`
- `TeamGameweekScoreRepository.findByFantasyTeam_IdAndGameweek_Id`,
`findByFantasyTeam_IdOrderByGameweek_NumberAsc`
- `GameweekTransferRepository.findByFantasyTeam_IdAndGameweek_IdOrderBySortOrderAsc`,
`findByFantasyTeam_IdOrderByGameweek_NumberAscSortOrderAsc`,
`deleteByFantasyTeam_IdAndGameweek_Id`

DTOs (records)

- **SnapshotRequest** (+ nested competition/gameweek/team/player/pick) — ingest body.
- **TransfersRequest** (+ round + pair) — ingest transfers; in/out may be null for LaLiga.
- **HomePageResponse** — `defaultCompetitionId` / `defaultCompetitionSlug`, `LeagueBrand` list, `PlayerOfTheWeek` list.
- **CompetitionResponse** — list item including branding URLs/colours.
- **TeamViewResponse** — dashboard squad: picks (`injured` / `suspended`), transfers, transfer-market summary (`mostExpensivePurchase` / `highestSale` highlights), captain display fields (`points`, `basePoints`, `captainMultiplier`).
- **CompareResponse** / **PlayerDelta**
- **TotalsResponse** / **GameweekTotal**

Tests

- **ApiSliceTest** — MockMvc slice: SofaScore chips, LaLiga market + transfer highlights, FPL badges, WSL branding (tournament 1044), captain display, injured/suspended flags.
- **BackendApplicationTests** — context load.

Frontend (Angular 21)

Standalone components, signals, `HttpClient`. Routes: homepage + leagues dashboard. GSAP is used only for the homepage POTW marquee.

Bootstrap

- **main.ts** — `bootstrapApplication(App, appConfig)`.
- **appConfig** — router, HTTP, global error listeners.
- **routes** (**app.routes.ts**) — `'` → `HomeComponent`; `leagues` → `DashboardComponent`; wildcard → `'`.
- **App** — shell with `<router-outlet>`.

HomeService

Thin GET client at `http://localhost:8080/api/v1/home`.

- **getHome()** — returns `HomePage`.

CompetitionService

Thin GET client at `http://localhost:8080/api/v1/competitions`. Created so the UI never POSTs.

- **getCompetitions**
- **getTeam(id, gameweek?)**
- **getGameweek(id, n)** — available but dashboard uses `getTeam`.
- **compare / totals**

tracker.ts models

TypeScript mirrors of GET JSON: `HomePage`, `LeagueBrand`, `PlayerOfTheWeek`, `Competition`, `TeamView`, `PickView` (injured / suspended), `TransferView`, `TransferHighlight`, `TransferMarket*` (incl. season extremes), `CompareView`, `PlayerDelta`, `TotalsView`.

HomeComponent — landing page

Why: brand-first homepage with league wash, MVP card, and a sliding Players of the Week rail.

Signals: `home`, `error`, `loading`.

Computed

- **colorMix** — builds `--home-mix` gradient from every `leagueBrands` primary/secondary pair.
- **leaguesHref / leaguesQuery** — CTA to `/leagues` with optional `?competition=` from `defaultCompetitionId`.
- **topPlayerOfTheWeek** — cross-competition MVP (max `points` in `playersOfTheWeek`).
- **marqueePlayers** — `[...players, ...players]` so GSAP can loop seamlessly on half the track width.

Lifecycle / marquee

- **constructor** — `HomeService.getHome()`; `afterRenderEffect` calls `setupMarquee` when `#marqueeTrack` exists; `DestroyRef` → `killMarquee`.
- **setupMarquee(track)** — GSAP `fromTo x: 0 → -scrollWidth/2`, `ease none`, `repeat: -1`, `~28 px/s`; skips rebuild when distance unchanged; preserves pause state.
- **killMarquee** — kills tween; clears transform.
- **pauseMarquee / resumeMarquee** — bound to marquee `mouseenter / mouseleave`.

Helpers: `hideImage`, `shirtLabel`, `isTopPlayer` (MVP highlight on duplicated marquee cards).

Homepage POTW / MVP styling (`home.css`)

Class / token	Purpose
<code>.home</code> , <code>.home-wash</code>	Full-page atmosphere; wash uses <code>--home-mix</code> + soft radials.
<code>.home-logos</code> , <code>.home-logo</code> , <code>.home-logo-0...-10</code>	Floating competition logos; one unique slot each (no <code>i % 5</code> overlap).
<code>.home-content</code> , <code>.home-hero</code> , <code>.eyebrow</code> , <code>.lede</code> , <code>.cta</code> , <code>.banner</code>	Hero copy + CTA.
<code>.mvp</code> , <code>.mvp-badge</code> , <code>.mvp-badge-compact</code>	Featured Top player of the week + MVP chip.
<code>.potw</code> , <code>.potw-head</code>	Section title for the rail.
<code>.potw-marquee</code>	Overflow hidden + edge fade mask.
<code>.potw-rail</code>	Flex <code>width: max-content</code> track GSAP translates.
<code>.potw-card</code>	Card chrome; <code>--potw-primary / --potw-secondary</code> from API.

Class / token	Purpose
<code>.potw-card-mvp</code>	Lime border + stronger glow for the MVP duplicate.
<code>.potw-meta</code> , <code>.comp-logo</code> , <code>.comp-name</code>	Competition row on the card.
<code>.portrait-stage</code> , <code>.portrait</code> , <code>.portrait-fallback</code> , <code>.crest</code>	Portrait plane + club crest overlay.
<code>.player-copy</code> , <code>.shirt</code> , <code>.name</code> , <code>.club</code> , <code>.gw-label</code>	Shirt, name, club, Round N • X pts.

Template: `home.html` — wash, logo field, hero, MVP `ng-template` card, marquee of cards sharing `#playerCard`.

DashboardComponent — leagues page

Why: one page for every league — branding, XI, bench/squad, GW standout card, status chips, season high/low weeks, transfers (LaLiga deal cards), compare, totals.

Signals: `competitions`, `selectedId`, `gameweek`, `fromGw`, `toGw`, `teamView`, `compareView`, `totalsView`, `error`, `loading`.

Computed

- `selectedCompetition` — current dropdown row.
- `pageBrand` — CSS variables from API primary/secondary + logo (page background).
- `starters / bench` — split on `role === 'starter'` (LaLiga “squad” still uses non-starter role).
- `isOfficialLaLiga / isOfficialFpl` — copy and FPL neon classes.
- `reserveHeading` — “Squad” vs “Bench”.
- `showTransferCounterpart` — LaLiga or any move with a counterpart.
- `playerOfTheWeek` — max `pick.points` in the current `teamView` (GW card).
- `highestScoringGameweek / lowestScoringGameweek` — season extremes from `totalsView.gameweeks`.
- `mostExpensivePurchase / highestPlayerSale` — Official LaLiga only; from `transferMarket` season highlights.

Methods

- `constructor` — loads competitions; honour `?competition=` query; selects default.
- `selectCompetition` — latest week, compare from first→latest, refresh all three GETs.
- `onCompetitionChange / onGameweekChange / onCompareChange` — template bindings.
- `transferLabel / priceFormat / counterpartLabel / hasMarketDeals` — LaLiga market wording and prices.
- `shirtLabel / transferHighlightShirt / transferHighlightMeta` — shirt / GW+counterpart for POTW-style transfer cards.
- `isInjured / isSuspended` — status chips on squad rows.
- `hideImage` — hide broken crest/portrait.
- `signed` — `+n / n` for compare deltas.
- `refreshAll / loadTeam / loadCompare / loadTotals` — HTTP; errors set `error`.

Competition-page GW POTW card styling (`dashboard.css`)

Same visual language as homepage cards, scoped under the competition view:

Class	Purpose
<code>.gw-potw</code> , <code>.gw-potw-head</code>	Centered “Player of the week” block above the squad.
<code>.potw-card</code> (+ nested portrait / crest / copy)	Card using <code>--potw-primary</code> / <code>--potw-secondary</code> from the selected competition.
<code>.potw-meta</code> , <code>.potw-comp-logo</code> , <code>.potw-comp-name</code>	Competition header on the GW card.
<code>.chip-injured</code> / <code>.chip-suspended</code>	Amber / red outline chips next to player names.
<code>.score-extremes</code>	Two-column Season-total-style cards for best/worst GW.
<code>.transfer-highlights</code> , <code>.transfer-highlights-grid</code>	Side-by-side LaLiga purchase/sale POTW cards in Transfers.

Also: `.page-brand`, `.page-brand-fpl`, `.panel-head-branded`, FPL neon overlays — competition header theming (not POTW-specific).

Templates/styles: `dashboard.html`, `dashboard.css`.

Python collector

Package `collector`, CLI `python -m collector`. Loads repo-root then `collector/.env`. Raw JSON goes to `collector/cache/<slug>/` (gitignored).

CLI (`__main__.py`)

- `main` — subcommands `import`, `import-excel`, `sync`, `pull`.
- `_run_import` — local JSON (squad or transfers export).
- `_run_import_excel` — official LaLiga workbook.
- `_run_sync` — legacy single-GW SofaScore URL.
- `_run_pull_command` — every SofaScore target in `.env`, plus FPL if `FPL_ENTRY_ID`, plus WSL if gameplay id + game token. `--competition` selects one slug (`premier-league-fantasy`, `nations-league`, `ws1-fantasy`, ...).

Canonical models (`models.py`)

- `PlayerPayload` / `PickPayload` / `Snapshot` — POST `/ingest/snapshots`. `PickPayload` carries `injured` / `suspended`. `Snapshot.to_dict` omits null team points.
- `TransferPlayer` / `TransferPair` / `TransferRound` / `TransfersBatch` — POST `/ingest/transfers`.
- `FantasyAdapter` — old interface (`fetch_competitions` / `fetch_squad` / `fetch_gameweek_scores`); SofaScore and file import still implement it. FPL/WSL use dedicated `pull_*` functions instead.

BackendPublisher

- **publish / publish_transfers** — POST with optional X-Ingest-Token. **BACKEND_URL** defaults to `http://localhost:8080`.

Pull orchestration (**pull.py**)

- **PullTarget** — slug + competition/squad/transfers/gameweek URLs.
- **pull_targets_from_env** — Premier League unprefixes **SOFAScore_*** or **SOFAScore_Premier_League_***; other leagues **SOFAScore_LaLiga_***, **Serie_A**, **Ligue_1**, **Bundesliga**, **Champions_League**, **Europa_League**, **Nations_League**, **MLS**, **Brasileirao**. A concrete `/round/{id}/squad` URL alone is enough for a first Nations League ingest.
- **complete_target** — from a Fantasy `.../competition/{id}` URL, derive `/transfers` and `/round/{roundId}/squad`.
- **_target_from_prefix** — `env` → **PullTarget**.
- **adapter_for_target** — **SofaScoreAdapter** for that target.
- **run_pull** — meta + rounds → **currentRound.id** (or `--gameweek` sequence) → squad → injury/suspension overlay → optional transfers → POST (unless `--dry-run`).
- **_save_json** — cache dump.

SofaScore Fantasy adapter (**adapters/sofascore.py**)

Why: replay **your** Network JSON URLs with your cookie; impersonate Chrome (**curl_cffi**) because Python **requests** was WAF 403.

- **CLUB_BY_ID / CLUB_BY_CODE** — expand abbreviations to full club names (PL includes Sunderland **41/SUN**, AFC Bournemouth **60/BOU**, Liverpool, Spurs, ...; plus LaLiga/Ligue 1/Serie A/MLS/Brasileirão). Transfers often only send **teamId** + **nameCode**.
- **resolve_club_name** — prefer **CLUB_BY_ID**, then **full API name**, then **CLUB_BY_CODE**. Full names win over codes so national **ESP** → Spain is not rewritten to Espanyol.
- **normalize_position**
- **snapshot_from_payload** — canonical snapshot, SofaScore squad, or “sofascore-like” shapes.
- **is_transfers_export / transfers_from_payload**
- **_from_canonical / _from_sofascore_squad / _from_sofascore_like**
- **_injured_from_sofascore / _first_unique_tournament / _normalize_season / _date_from_timestamp**
- **_transfer_player / _club_external_id / _optional_float / _optional_int**
- **SofaScoreSessionError** — 401 → refresh **SOFAScore_SESSION**.
- **SofaScoreAdapter**
 - **fetch_competitions / fetch_meta / fetch_rounds / fetch_squad / fetch_transfers / fetch_gameweek_scores**
 - **_get_json / _require_round_id**
- **round_id_from_meta** — current/next/previous or **userRounds** calendar by sequence.
- **_browser_headers / _http_get / _env_or / _normalize_session_cookie / _format_url** — strip `Cookie:` prefix; rewrite `/round/{digits}/`.
- **load_snapshot_file**

Round overlay (**adapters/sofascore_round.py**)

Uses **known** SofaScore paths only: `unique-tournament/{id}/seasons, .../season/{id}/players, .../events/round/{n}, event/{id}/lineups, team/{id}/players`.

- `current_season_id` — first season in the list (current).
- `overlay_from_lineups` — ratings + **injured** + **suspended** ids from `missingPlayers` (skips doubtful). Suspension wins over injury when both could match.
- `fetch_round_overlay` — all events in a round (`injuredIds` / `suspendedIds`).
- `fetch_injured_player_ids` — live `injury.status == out` (upcoming/live weeks).
- `apply_round_overlay` / `annotate_snapshot_injuries` — also sets `pick.suspended` (clears injured when suspended).
- `_is_suspended_missing` — SofaScore reasons **3** (FA/improper conduct), **11** (yellow accumulation), **12** (yellow/red), **13** (red), plus description tokens (`suspension`, `unavail`, ...).
- `_is_injured_missing` — injury-like missing rows that are **not** suspensions (reason **1** / `"injur..."`).
- `_injured_for_team` / `_player_id` / `_optional_int`

Player directory (`adapters/sofascore_directory.py`)

Why: official FPL/WSL ids are not SofaScore ids; portraits (and FPL display names) need numeric player ids / SofaScore `playerName`.

- `fold_name` / `club_key` — accents, FC/Women/Lionesses, Brighton & Hove, Man Utd aliases.
- `fetch_tournament_players` / `fetch_premier_league_players`
- `resolve_sofascore_player` — club then name tokens.
- `search_sofascore_player` — `search/all` fallback.
- `_tokens_match`

Tournament constants: PL **17**, WSL **1044**, WSL2 **10553**. Nations League branding tournament id **10783**.

Official FPL (`adapters/fpl.py`)

Public `fantasy.premierleague.com/api`. No cookie.

- `fpl_entry_id` / `pull_fpl` — bootstrap, entry, history, transfers, per-GW picks + live; SofaScore overlay on tournament 17.
- `catalog_from_bootstrap` / `competition_payload` / `team_payload` / `snapshot_from_picks` / `transfers_from_fpl`
- `_pick_from_fpl` / `_transfer_pair` / `_transfer_player` / `_player_from_element`
- `_display_name` — prefer SofaScore `playerName`, then FPL `first_name` + `second_name`, then `web_name` (dashboard shows full names, not "Virgil" / "Gabriel").
- `_resolve_sofascore_player` / `_sofascore_player_id` — cache hit rows for id + name; legacy id cache kept for injury overlays.
- `_optional_shirt` / `_safe_round_overlay` / `_current_injured_ids`
- `_event_status` / `_season` / `_price` — prices in tenths of a million.
- `_optional_float` / `_get_json` / `_save_json`

Captain live points are **raw**; dashboard ×2.

Official WSL (`adapters/wsl.py`)

Auth is request header **x-game-token** (env **WSL_GAME_TOKEN**, fallback **WSL_BEARER**). Auth0 Bearer is identity, not my-team.

- **wsl_game_token / wsl_gameplay_id / wsl_configured / pull_wsl**
- **competition_payload / team_payload / snapshot_from_my_team / transfers_from_squads** — transfers are week-to-week **arrTeam** diffs.
- **_pick_from_wsl / _player_from_row / _sofascore_player_id / _transfer_player**
- **_current_injured_ids / _safe_round_overlay / _safe_wsl_players** — overlay 1044 + 10553.
- **_gameweek_status / _starts_at / _season**
- **_raw_points** — my-team **totalPoints** **already includes captain ×2**; store half so the API does not double again.
- **_team_points / _position_name / _short_id / _index / _optional_float / _feed_value / _get_json / _save_json**

Official LaLiga Excel (**adapters/excel_laliga.py**)

App-only game (100M€ market, XI + unused squad, unpaired buys/sells, no captains).

- **load_workbook_payloads** — snapshots + transfer batch.
- **_competition_and_team / _snapshots_from_squads / _transfers_from_sheet**
- **_pick_from_row / _deal_from_row / _transfer_player**
- **_key_values / _table_rows / _id_or_slug / _slug / _digits / _int / _optional_float / _optional_text / _position / _action / _injured / _role**

File import (**adapters/file_import.py**)

- **FileImportAdapter.fetch_competitions / fetch_squad / fetch_gameweek_scores / _load** — saved Network JSON when URLs are unavailable.

Excel template builder (**templates/build_laliga_fantasy_workbook.py**)

- **build** — writes **laliga-fantasy-oficial.xlsx**.
- **_readme / _competition / _squads / _transfers / _clubs / _header_row / _fill_row / _list**

Collector tests

test_fpl, test_wsl, test_pull (includes Nations League squad-only target), **test_sofascore_***, **test_excel_laliga, test_sofascore_directory** — mapping and env targeting without printing secrets.

Environment (gitignored)

Variable	Purpose
SOFASCORE_SESSION	Cookie value from a Fantasy XHR
SOFASCORE_* / SOFASCORE_<LEAGUE>_*	Competition / squad / transfers URLs (incl. SOFASCORE_NATIONS_LEAGUE_*)
FPL_ENTRY_ID	Public FPL team id

Variable	Purpose
WSL_GAMEPLAY_ID	Gameplay UUID in my-team path
WSL_GAME_TOKEN (or WSL_BEARER)	x-game-token header (~24h). Not Auth0.
BACKEND_URL / INGEST_TOKEN	Publisher target

Never log tokens. HTTP 401 on SofaScore → refresh cookie. HTTP 401 on WSL → copy a fresh x-game-token.

Docs / screenshots

- docs/TECHNICAL.md — this file.
- docs/screenshots/ — README media (home.png, leagues.png, potw-marquee.gif, squad-status-badges.png, score-extremes.png, laliga-transfer-highlights.png).